

## Database Schema Documentation

# A Mobile Application for Facility Management

### Team

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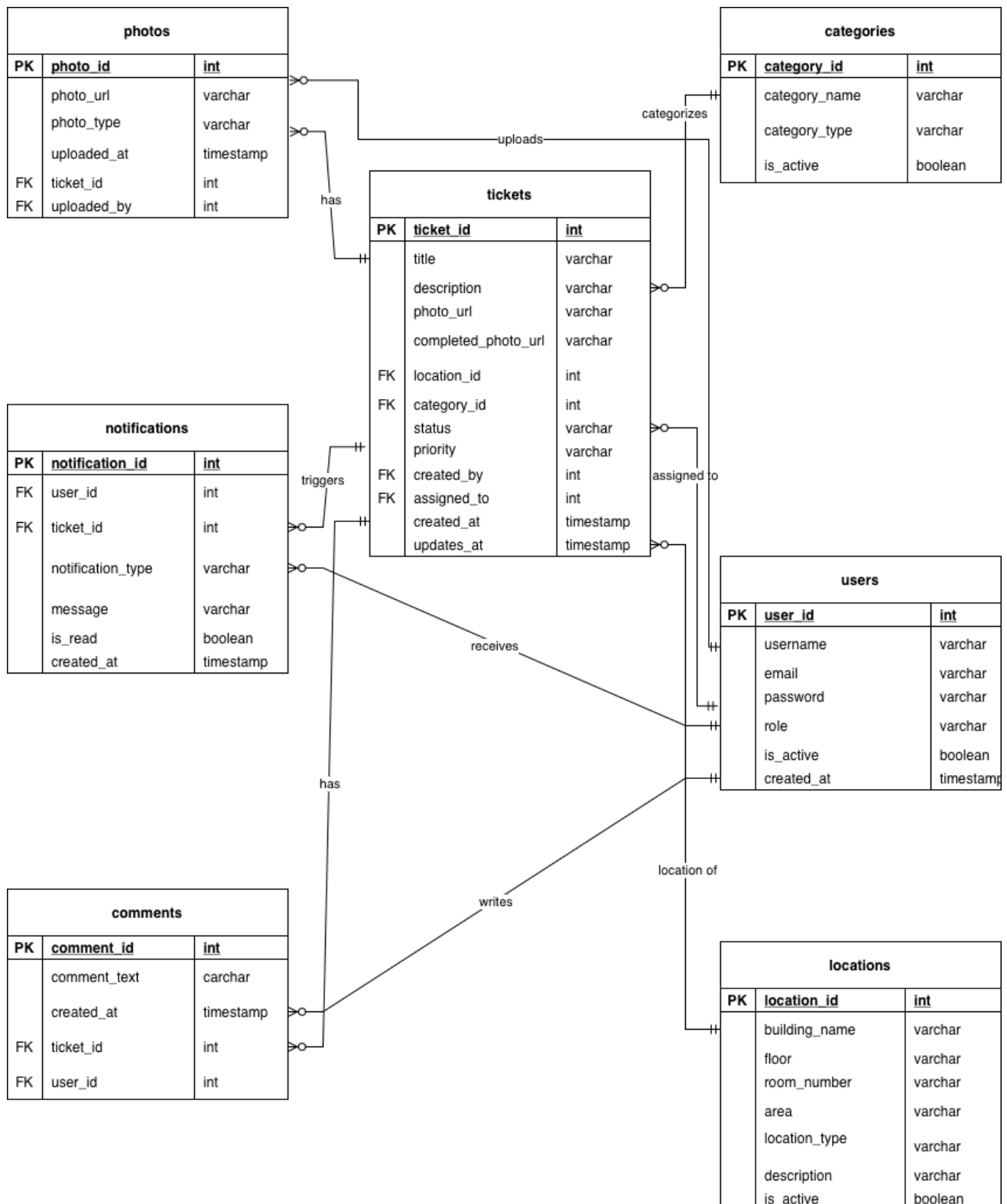
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### Tutorial Details

T4 | Hassan Abdelqader

## Entity-Relationship Diagram



## Table Definitions

### USERS

| Column     | Type         | Constraints                                                           |
|------------|--------------|-----------------------------------------------------------------------|
| user_id    | SERIAL       | PK                                                                    |
| username   | VARCHAR(100) | NOT NULL, UNIQUE                                                      |
| email      | VARCHAR(150) | NOT NULL, UNIQUE                                                      |
| password   | VARCHAR(255) | NOT NULL                                                              |
| role       | VARCHAR(30)  | NOT NULL, CHECK: community_member / facility_manager / worker / admin |
| is_active  | BOOLEAN      | DEFAULT TRUE                                                          |
| created_at | TIMESTAMP    | DEFAULT NOW()                                                         |

### LOCATIONS

| Column        | Type         | Constraints             |
|---------------|--------------|-------------------------|
| location_id   | SERIAL       | PK                      |
| building_name | VARCHAR(100) | NOT NULL                |
| floor         | VARCHAR(20)  | —                       |
| room_number   | VARCHAR(50)  | —                       |
| area          | VARCHAR(100) | —                       |
| location_type | VARCHAR(20)  | CHECK: indoor / outdoor |
| description   | VARCHAR(255) | —                       |
| is_active     | BOOLEAN      | DEFAULT TRUE            |

**CATEGORIES**

| Column        | Type        | Constraints  |
|---------------|-------------|--------------|
| category_id   | SERIAL      | PK           |
| category_name | VARCHAR(50) | NOT NULL     |
| category_type | VARCHAR(50) | —            |
| is_active     | BOOLEAN     | DEFAULT TRUE |

**TICKETS**

| Column              | Type         | Constraints                                                                    |
|---------------------|--------------|--------------------------------------------------------------------------------|
| ticket_id           | SERIAL       | PK                                                                             |
| title               | VARCHAR(200) | NOT NULL                                                                       |
| description         | VARCHAR(500) | NOT NULL                                                                       |
| photo_url           | VARCHAR(500) | —                                                                              |
| completed_photo_url | VARCHAR(500) | —                                                                              |
| location_id         | INT          | FK → LOCATIONS                                                                 |
| category_id         | INT          | FK → CATEGORIES                                                                |
| status              | VARCHAR(20)  | DEFAULT 'pending', CHECK: pending / assigned / in_progress / resolved / closed |
| priority            | VARCHAR(20)  | DEFAULT 'medium', CHECK: low / medium / high / urgent                          |
| created_by          | INT          | FK → USERS                                                                     |
| assigned_to         | INT          | FK → USERS                                                                     |
| created_at          | TIMESTAMP    | DEFAULT NOW()                                                                  |
| updated_at          | TIMESTAMP    | DEFAULT NOW()                                                                  |

**COMMENTS**

| Column       | Type         | Constraints                   |
|--------------|--------------|-------------------------------|
| comment_id   | SERIAL       | PK                            |
| comment_text | VARCHAR(255) | NOT NULL                      |
| created_at   | TIMESTAMP    | DEFAULT NOW()                 |
| ticket_id    | INT          | FK → TICKETS (CASCADE DELETE) |
| user_id      | INT          | FK → USERS                    |

**PHOTOS**

| Column      | Type         | Constraints                   |
|-------------|--------------|-------------------------------|
| photo_id    | SERIAL       | PK                            |
| photo_url   | VARCHAR(500) | NOT NULL                      |
| photo_type  | VARCHAR(20)  | CHECK: issue / completion     |
| uploaded_at | TIMESTAMP    | DEFAULT NOW()                 |
| ticket_id   | INT          | FK → TICKETS (CASCADE DELETE) |
| uploaded_by | INT          | FK → USERS                    |

**NOTIFICATIONS**

| Column            | Type         | Constraints                                    |
|-------------------|--------------|------------------------------------------------|
| notification_id   | SERIAL       | PK                                             |
| user_id           | INT          | FK → USERS (CASCADE DELETE)                    |
| ticket_id         | INT          | FK → TICKETS (CASCADE DELETE)                  |
| notification_type | VARCHAR(30)  | CHECK: status_change / assignment / completion |
| message           | VARCHAR(255) | —                                              |
| is_read           | BOOLEAN      | DEFAULT FALSE                                  |
| created_at        | TIMESTAMP    | DEFAULT NOW()                                  |

## What each table does and why it exists:

**USERS** — Everyone who touches the system. The role column is what separates a community member (reports issues) from a worker (fixes them), a facility manager (oversees), and an admin (manages the whole platform). One table handles all four rather than splitting them out.

**LOCATIONS** — Where on campus the problem is. Broken down into building\_name, floor, room\_number, and area so you can get as specific as needed. The location\_type (indoor/outdoor) helps route the right kind of worker. is\_active lets you retire a building without deleting its history.

**CATEGORIES** — What kind of problem it is — electrical, plumbing, HVAC, etc. Kept in its own table so new categories can be added without touching any other structure. category\_type adds a second level of grouping if needed (e.g. grouping electrical and HVAC under "mechanical").

**TICKETS** — The heart of the system. Every reported issue lives here. It holds the full lifecycle of a problem: who reported it (created\_by), who's fixing it (assigned\_to), where it is (location\_id), what type it is (category\_id), how urgent it is (priority), and where it stands (status). The two photo URL columns — photo\_url for the reported issue and completed\_photo\_url for the finished repair — give before/after evidence directly on the ticket.

**COMMENTS** — The conversation thread on a ticket. Workers can update progress, managers can ask questions, members can add details — all tied to a specific ticket and user. Cascade delete means comments disappear cleanly when their ticket is removed.

**PHOTOS** — A more flexible photo store than the two URL columns on TICKETS. While TICKETS holds the primary before/after shots, this table supports multiple photos per ticket and tracks who uploaded each one and whether it's an issue photo or a completion photo. Useful when one repair needs several angles documented.

**NOTIFICATIONS** — The alerting layer. Whenever a ticket's status changes, gets assigned, or is completed, a row lands here for the relevant user. The is\_read flag drives the unread badge count in the app. Cascades on both USERS and TICKETS, so there's no orphaned notification data if either is deleted.

## Seed Data

```
INSERT INTO users (username, email, password, role, is_active)
VALUES
(
    'Mariam Student',
    'member@campuscare.com',
    '$2a$10$7EqJtq98hPqEX7fNZaFWoOHI7v3K7.Lf0ypAWR5jx5qj8YhZPKQhm',
    'community_member',
    TRUE
),
(
    'Omar Facility Manager',
    'manager@campuscare.com',
    '$2a$10$7EqJtq98hPqEX7fNZaFWoOHI7v3K7.Lf0ypAWR5jx5qj8YhZPKQhm',
    'facility_manager',
    TRUE
),
(
    'Ahmed Maintenance Worker',
    'worker@campuscare.com',
    '$2a$10$7EqJtq98hPqEX7fNZaFWoOHI7v3K7.Lf0ypAWR5jx5qj8YhZPKQhm',
    'worker',
    TRUE
),
(
    'CampusCare Admin',
    'admin@campuscare.com',
    '$2a$10$7EqJtq98hPqEX7fNZaFWoOHI7v3K7.Lf0ypAWR5jx5qj8YhZPKQhm',
    'admin',
    TRUE
);

INSERT INTO categories (category_name, category_type, is_active)
VALUES
('Electrical', 'Maintenance', TRUE),
('Plumbing', 'Maintenance', TRUE),
('Cleaning', 'Service', TRUE),
('Furniture', 'Maintenance', TRUE),
('Air Conditioning', 'Maintenance', TRUE),
('Safety', 'Urgent Maintenance', TRUE);

INSERT INTO locations (
    building_name,
    floor,
    room_number,
```

```

    area,
    location_type,
    description,
    is_active
)
VALUES
('Building A', '1', 'A101', 'North Wing', 'indoor', 'Lecture Hall A101', TRUE),
('Building B', '2', 'B205', 'South Wing', 'indoor', 'Computer Lab B205', TRUE),
('Library', 'Ground', 'L-G01', 'Main Reading Area', 'indoor', 'Main library study area',
TRUE),
('Cafeteria', 'Ground', 'C-G01', 'Food Court', 'indoor', 'Main cafeteria entrance', TRUE),
('Parking Area', NULL, 'P1', 'East Side', 'outdoor', 'Student parking area', TRUE),
('Building C', '3', 'C303', 'West Wing', 'indoor', 'Seminar room C303', TRUE);

INSERT INTO tickets (
    title,
    description,
    photo_url,
    completed_photo_url,
    location_id,
    category_id,
    status,
    priority,
    created_by,
    assigned_to,
    created_at,
    updated_at
)
VALUES
(
    'Broken chair in lecture hall',
    'One of the chairs in Lecture Hall A101 is broken and unsafe for students to use.',
    'https://res.cloudinary.com/demo/image/upload/v1690000000/samples/chair.jpg',
    NULL,
    1,
    4,
    'pending',
    'medium',
    1,
    NULL,
    NOW() - INTERVAL '4 days',
    NOW() - INTERVAL '4 days'
),
(
    'Flickering light in computer lab',
    'The ceiling light in Computer Lab B205 keeps flickering during lectures.',

```



```

    'https://res.cloudinary.com/demo/image/upload/v1690000000/samples/light.jpg',
    NULL,
    2,
    1,
    'assigned',
    'high',
    1,
    3,
    NOW() - INTERVAL '3 days',
    NOW() - INTERVAL '2 days'
),
(
    'Water leak near cafeteria entrance',
    'There is water leaking near the cafeteria entrance and the floor is becoming
slippery.',
    'https://res.cloudinary.com/demo/image/upload/v1690000000/samples/water.jpg',
    NULL,
    4,
    2,
    'in_progress',
    'urgent',
    1,
    3,
    NOW() - INTERVAL '2 days',
    NOW() - INTERVAL '1 day'
),
(
    'Overflowing trash bin in library',
    'The trash bin in the main library study area is full and needs cleaning.',
    'https://res.cloudinary.com/demo/image/upload/v1690000000/samples/people/bicycle.jpg',
    'https://res.cloudinary.com/demo/image/upload/v1690000000/samples/landscapes/nature-mounta
ins.jpg',
    3,
    3,
    'resolved',
    'low',
    1,
    3,
    NOW() - INTERVAL '1 day',
    NOW() - INTERVAL '3 hours'
),
(
    'AC not cooling in seminar room',
    'The air conditioner in room C303 is running but the room is still very hot.',

```

```

'https://res.cloudinary.com/demo/image/upload/v1690000000/samples/ecommerce/leather-bag-gray.jpg',
    NULL,
    6,
    5,
    'pending',
    'high',
    1,
    NULL,
    NOW() - INTERVAL '8 hours',
    NOW() - INTERVAL '8 hours'
);

INSERT INTO comments (comment_text, ticket_id, user_id)
VALUES
('I have received the assignment and will inspect the light today.', 2, 3),
('The leak area has been checked. Repair is currently in progress.', 3, 3),
('Cleaning completed. The library area is now clean.', 4, 3);

INSERT INTO photos (photo_url, photo_type, ticket_id, uploaded_by)
VALUES
('https://res.cloudinary.com/demo/image/upload/v1690000000/samples/chair.jpg', 'issue', 1, 1),
('https://res.cloudinary.com/demo/image/upload/v1690000000/samples/light.jpg', 'issue', 2, 1),
('https://res.cloudinary.com/demo/image/upload/v1690000000/samples/water.jpg', 'issue', 3, 1),
('https://res.cloudinary.com/demo/image/upload/v1690000000/samples/people/bicycle.jpg', 'issue', 4, 1),
('https://res.cloudinary.com/demo/image/upload/v1690000000/samples/landscapes/nature-mountains.jpg', 'completion', 4, 3);

INSERT INTO notifications (
    user_id,
    ticket_id,
    notification_type,
    message,
    is_read,
    created_at
)
VALUES
(2, 1, 'status_change', 'New issue submitted: Broken chair in lecture hall', FALSE, NOW() - INTERVAL '4 days'),
(3, 2, 'assignment', 'You have been assigned a new issue: Flickering light in computer lab', FALSE, NOW() - INTERVAL '2 days'),

```

```
(1, 2, 'status_change', 'Your issue "Flickering light in computer lab" has been assigned  
to a worker', FALSE, NOW() - INTERVAL '2 days'),  
(2, 3, 'status_change', 'Worker updated issue "Water leak near cafeteria entrance"',  
FALSE, NOW() - INTERVAL '1 day'),  
(2, 4, 'completion', 'Worker uploaded a completion photo for issue "Overflowing trash bin  
in library"', FALSE, NOW() - INTERVAL '3 hours'),  
(1, 4, 'status_change', 'Your issue "Overflowing trash bin in library" has been marked as  
resolved', FALSE, NOW() - INTERVAL '3 hours');
```